Assessing China's Nuclear Capability

A Python-Based Open-Source Strategic Forces Study

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About this project
Built with Python, pandas, and matplotlib. Uses open-source public estimates only. Designed as a research and portfolio project covering stockpile estimates, delivery-system modernization, global comparison, forecast scenarios, proxy-based spending analysis, and budget-context interpretation.

Important caveat: This report uses open-source estimates only. It does not use classified information and does not provide operational targeting, weapons design, or tactical employment guidance.

Executive Summary

This project analyzes China's current nuclear capability using open-source estimates from public research organizations, government reporting, and defense-budget data. The study focuses on stockpile growth, delivery-system modernization, global comparison, transparent forecast scenarios, proxy-based spending analysis, and budget-context interpretation.

The core finding is that China has moved from a historically smaller minimum-deterrent posture toward a larger and more diversified nuclear force. Public estimates place China's arsenal at about 600 nuclear warheads in 2025, with U.S. Department of Defense reporting assessing that China remains on track to exceed 1,000 warheads by 2030. This remains far below U.S. and Russian total arsenals, but it represents a major shift in the global nuclear balance.

The spending analysis adds a finance-oriented dimension. Because China does not publish a dedicated nuclear weapons budget, this project uses proxy indicators: official defense budgets, estimated actual military expenditure, and ICAN-style nuclear weapons spending estimates. Based on the current dataset, China spent an estimated $12.5 billion on nuclear weapons in 2024, equal to roughly 3.93% of estimated actual military spending and about $25.0 million per estimated warhead inventory.

Research Questions

• How has China's nuclear capability changed over the last decade, and what do current open-source estimates suggest about its future role in global nuclear deterrence?

• Is China's nuclear expansion visible in public spending estimates, and how does estimated nuclear weapons spending compare with stockpile growth?

Methodology

• Open-source estimates of Chinese nuclear warheads from 2015 to 2025

• Public global nuclear force estimates for comparison

• A qualitative delivery-system summary covering land, sea, and air components

• Low, base, and high scenario projections from 2025 to 2035

• Proxy-based spending indicators, including official defense budgets, estimated actual military spending, and estimated nuclear weapons spending

• Budget-context sources that help explain why official Chinese fiscal and defense figures require careful interpretation

Official Chinese budget documents provide a useful national defense budget baseline, but they do not identify a separate nuclear weapons budget. Therefore, this study treats official defense spending as a broad proxy rather than a direct nuclear-program measure.

China's broader fiscal context is also relevant because public budget figures are distributed across multiple budget layers. This makes headline budget figures useful but incomplete indicators of the state's fiscal stance.

Key Findings

• China's estimated stockpile has grown quickly, reaching roughly 600 warheads in 2025. SIPRI and FAS/Bulletin estimates place China at about 600 warheads, while U.S. Department of Defense reporting assesses continued growth toward more than 1,000 warheads by 2030.

• China is developing a more complete nuclear triad. Public reporting describes land-based modernization, sea-based deterrent development, and some bomber-related nuclear roles.

• China remains below the United States and Russia in total estimated warheads, but it is becoming the third major nuclear power by estimated stockpile size.

• Spending analysis strengthens the project by connecting defense finance proxies with nuclear modernization indicators.

• Official budget figures require careful interpretation because the public defense budget is a useful baseline but does not necessarily capture all defense-related spending.

• Forecasts depend heavily on assumptions. The base scenario reaches roughly 1,000 warheads by 2030, while the high scenario reaches about 1,500 by 2035. These are illustrative scenarios rather than predictions.

Strategic and Fiscal Context

China's nuclear modernization should be viewed within a broader strategic and fiscal context. The project uses official Chinese budget reporting, U.S. Department of Defense assessments, SIPRI military expenditure estimates, and outside budget-analysis sources to distinguish between three different concepts:

• Official defense budget - the public budget figure reported by the Chinese government.

• Estimated actual military spending - outside estimates that attempt to capture broader military-related spending.

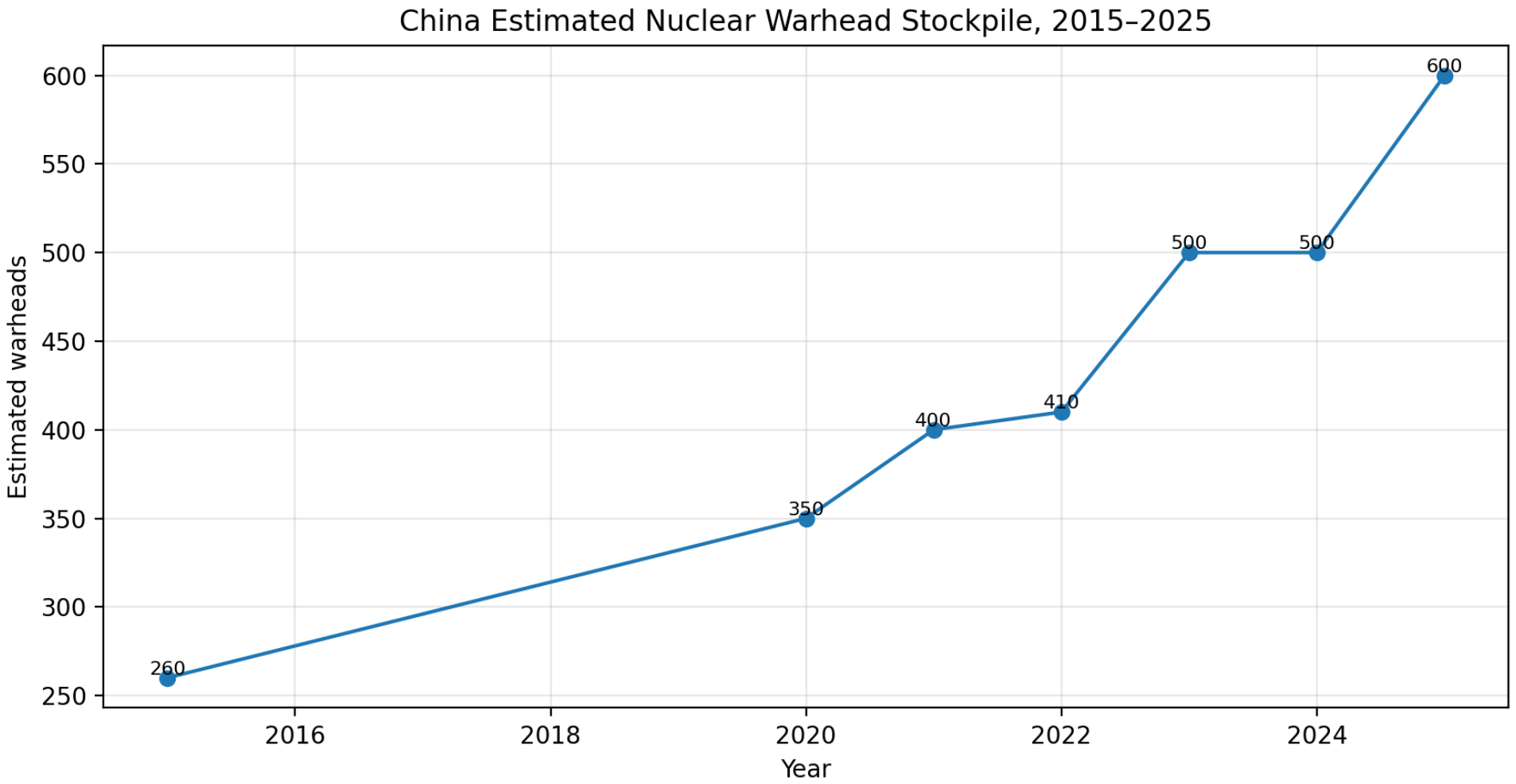
• Estimated nuclear weapons spending - outside estimates of nuclear weapons-related spending, which are more uncertain because China does not publish a dedicated nuclear weapons budget.

This distinction is important because a nuclear-weapons program can involve warheads, delivery systems, submarines, missiles, command and control, infrastructure, research, production facilities, and personnel. A single spending number cannot fully capture these categories.

The project also separates nuclear force-structure analysis from nuclear and missile proliferation analysis. CRS reporting shows that U.S. concerns about China also include export-control enforcement, China-based entities, and proliferation-sensitive transfers involving Iran, North Korea, and Pakistan. Those issues are important for broader strategic context, but they are not the primary focus of this stockpile and spending model.

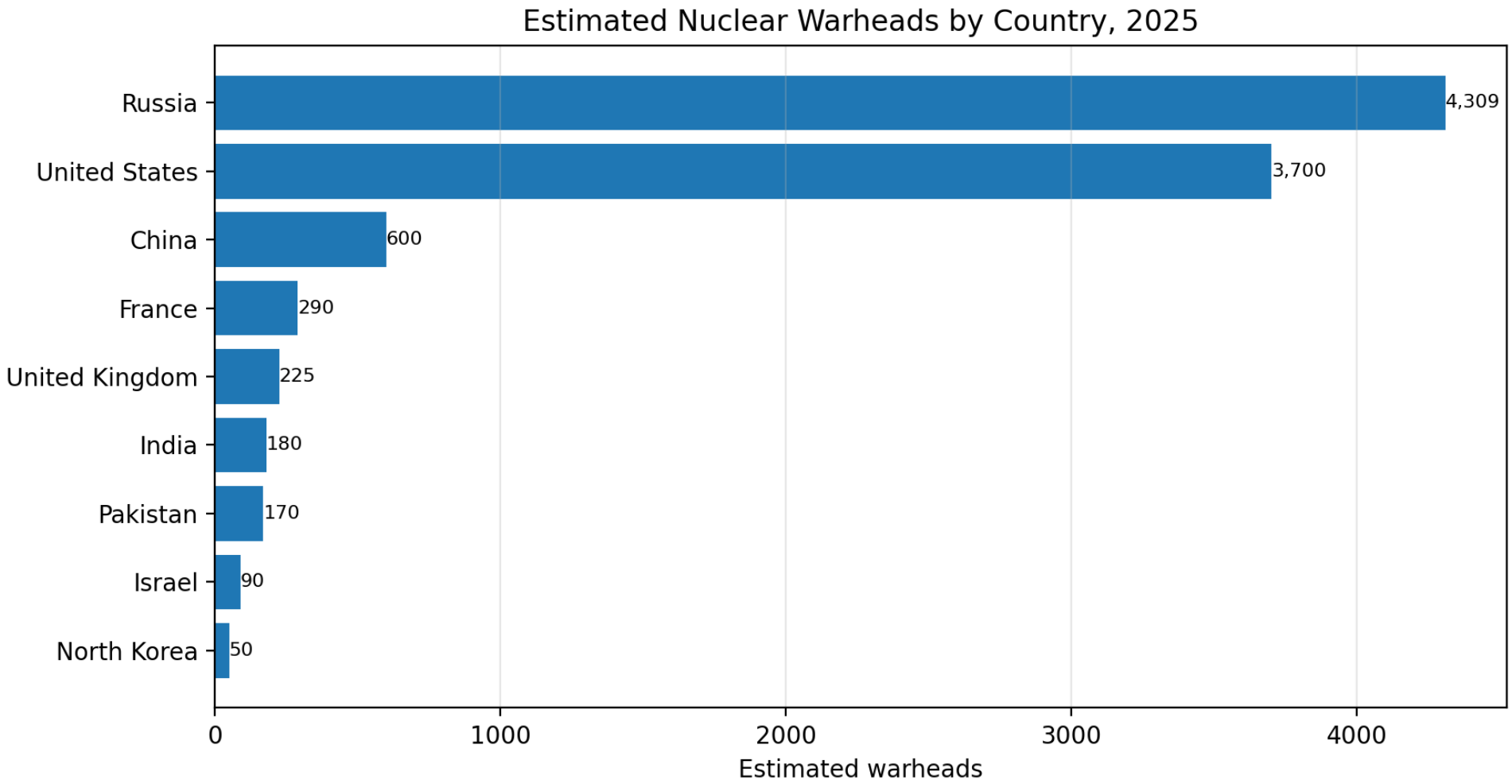
Generated Charts

The following figures were generated by the Python analysis pipeline and embedded for easier review. Chart notes are included directly in the figure images where relevant.



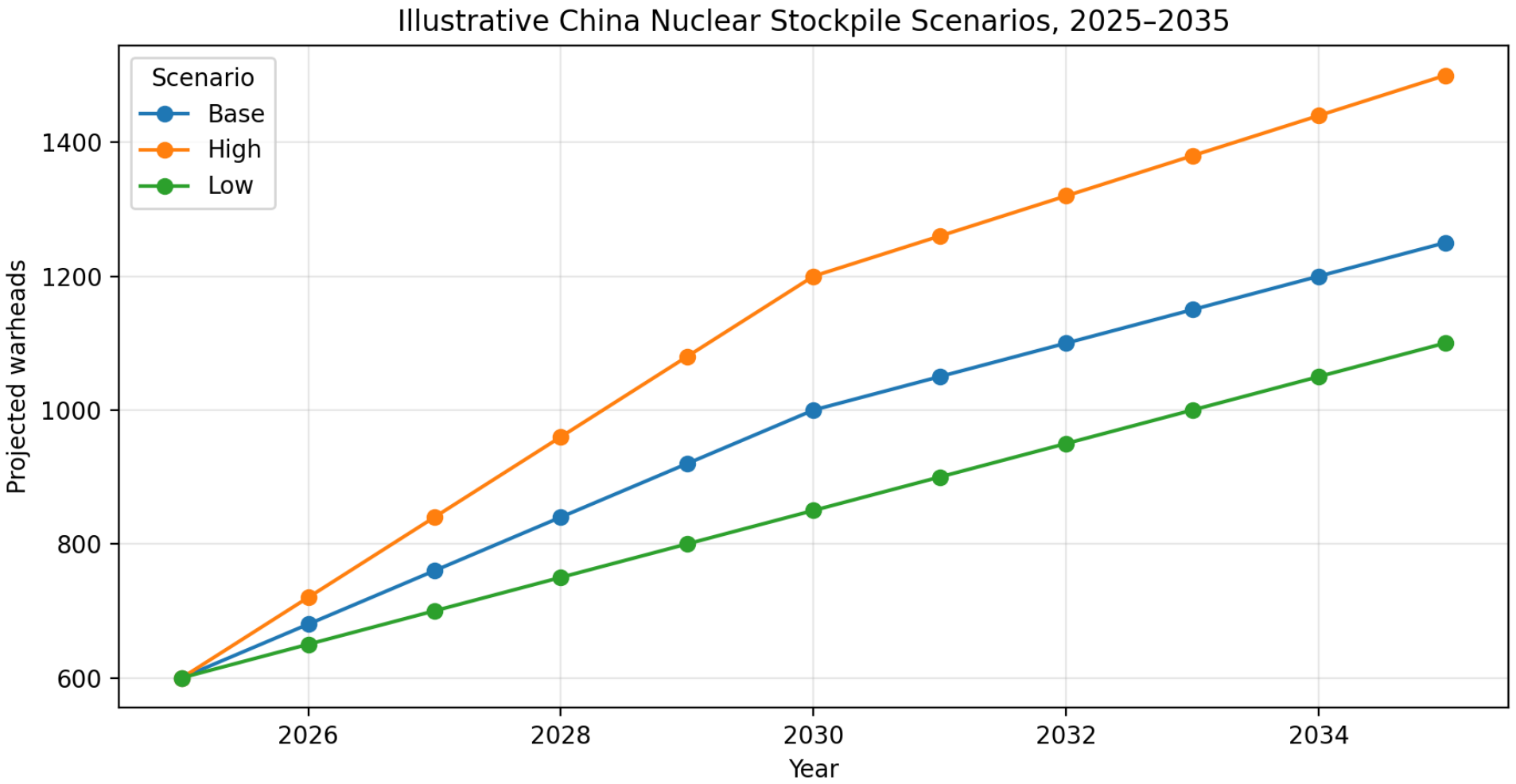
Note: Warhead totals are public estimates and should be interpreted as approximate.

Figure 1. China estimated nuclear warhead stockpile, 2015-2025



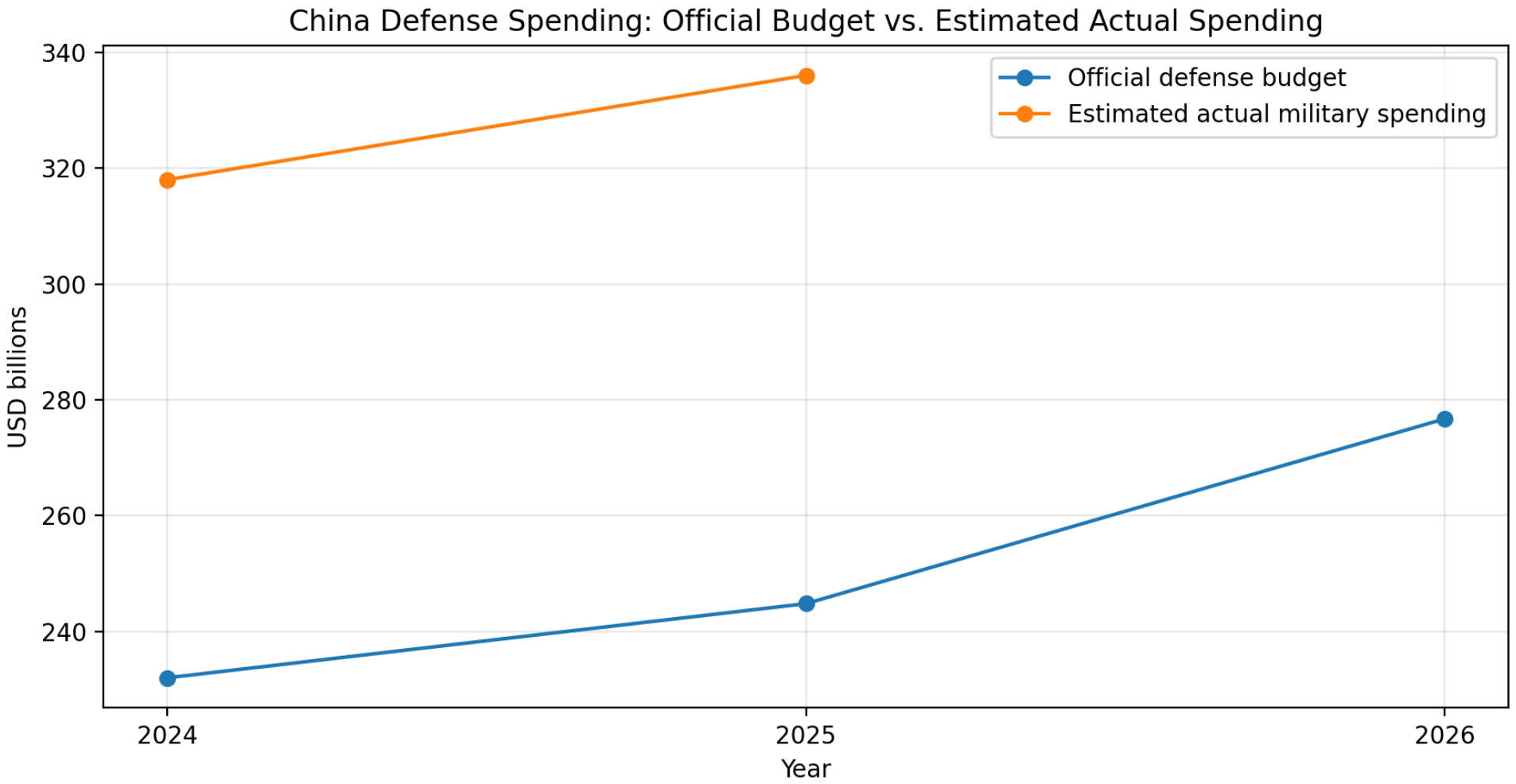
Note: Estimates are rounded and vary by source; some states maintain nuclear opacity.

Figure 2. Estimated nuclear warheads by country, 2025



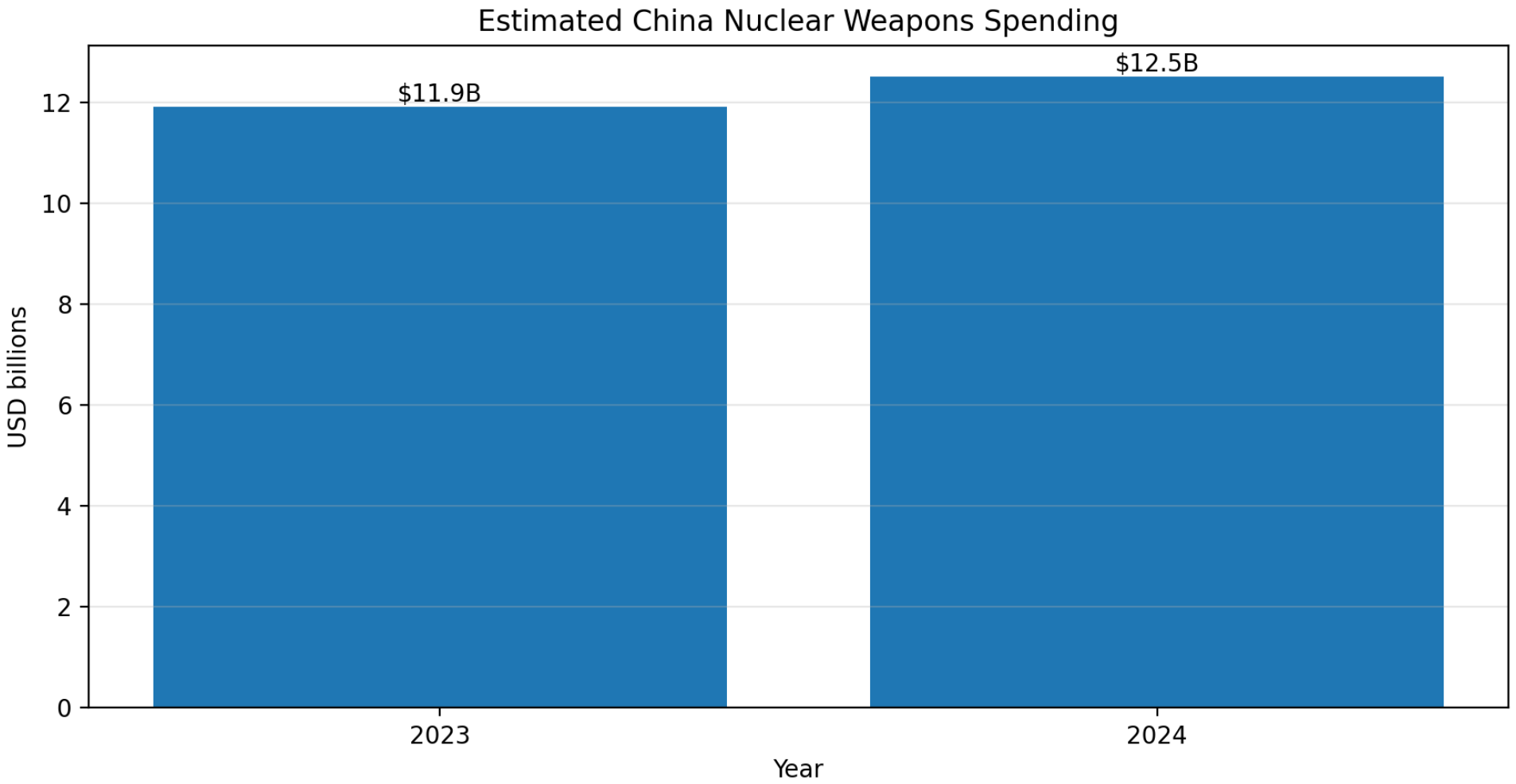
Note: Scenarios are illustrative projections, not intelligence predictions; the base case broadly aligns with public DoD 2030 assessments.

Figure 3. Illustrative China nuclear stockpile scenarios, 2025-2035



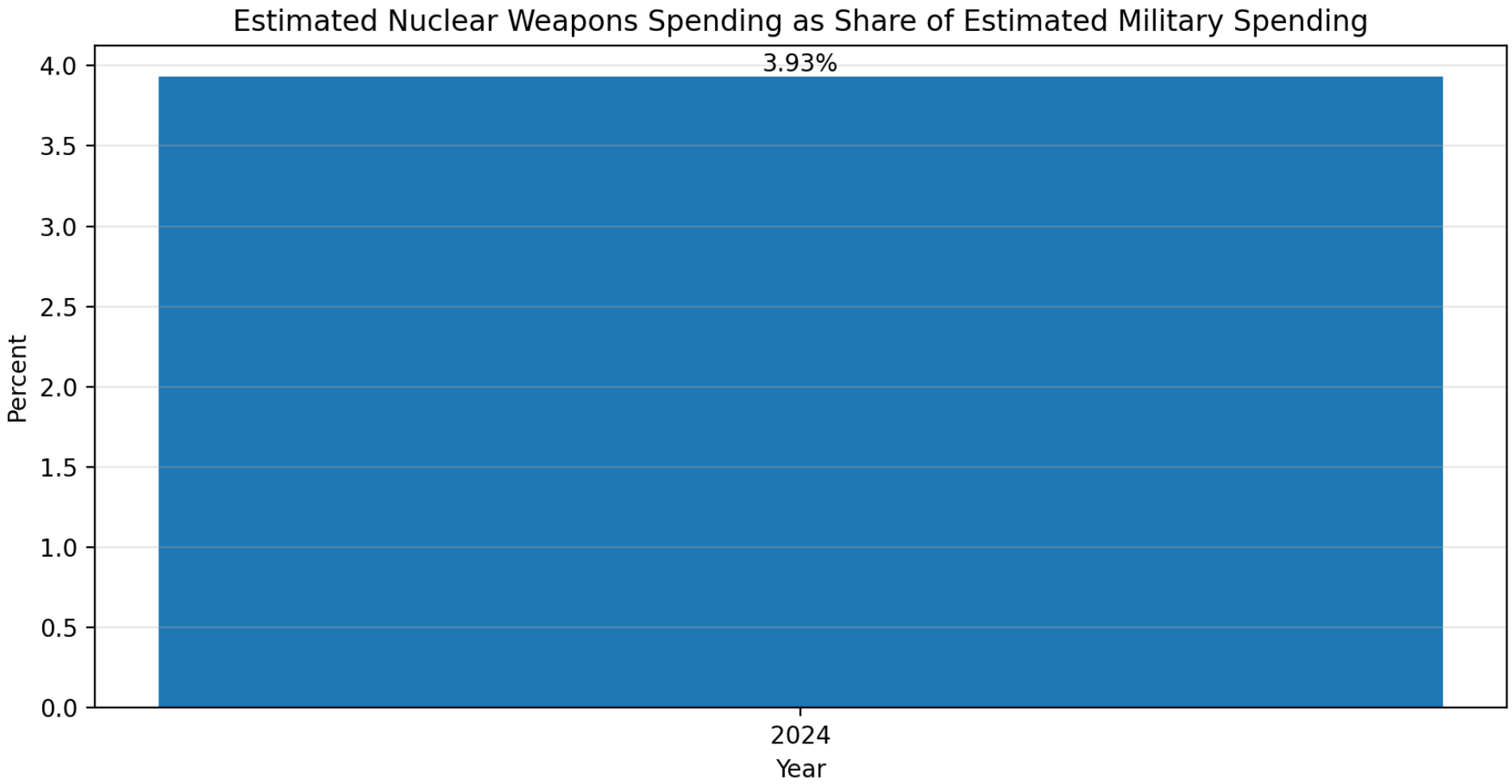
Note: Official defense budgets may not capture total military-related spending; estimated actual spending uses outside public estimates.

Figure 4. China defense spending: official budget vs. estimated actual spending



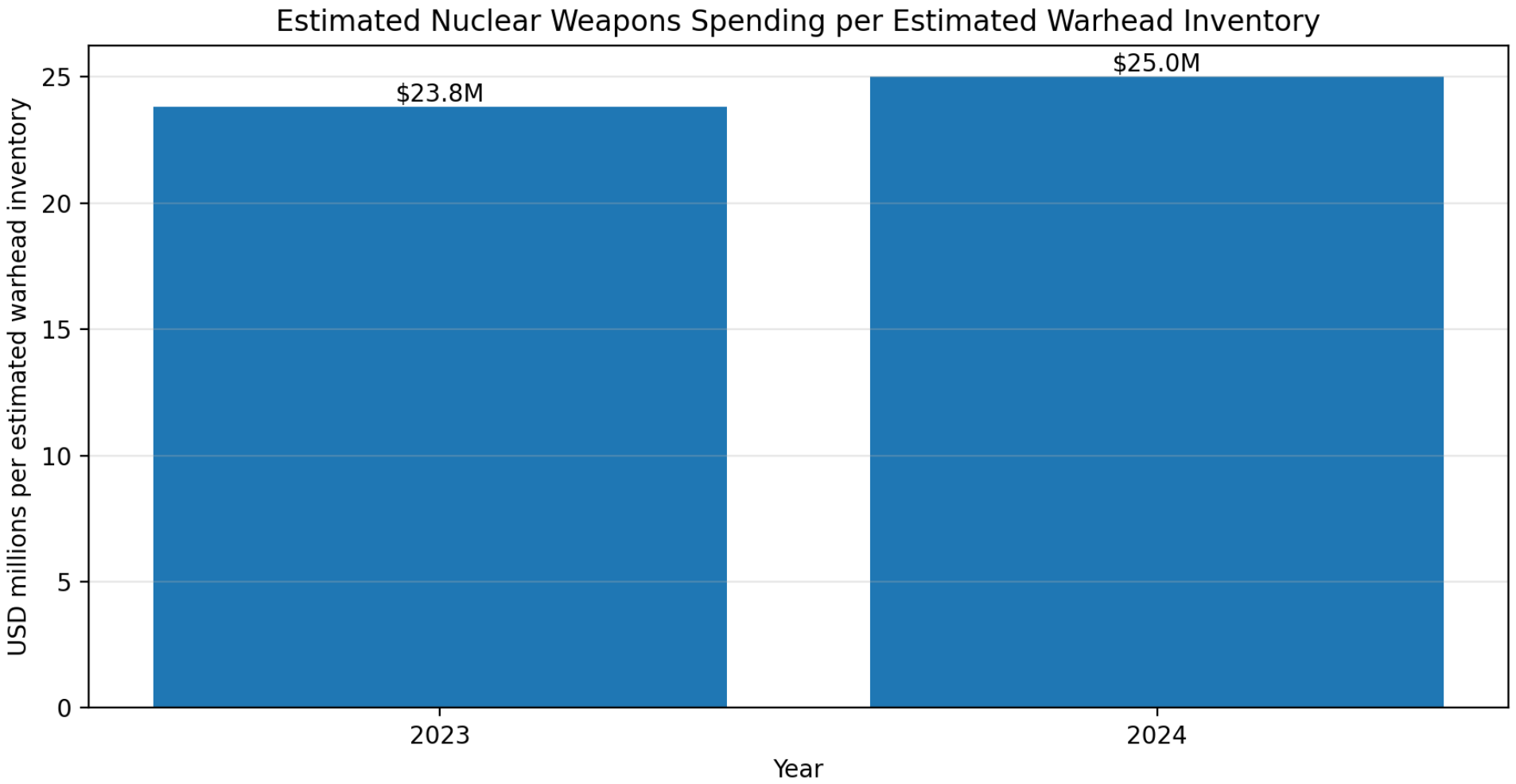
Note: China does not publish a separate nuclear weapons budget; figures are public estimates.

Figure 5. Estimated China nuclear weapons spending



Note: Share uses estimated nuclear weapons spending divided by estimated actual military spending.

Figure 6. Estimated nuclear weapons spending as share of estimated military spending



Note: This is not the cost of one warhead; it divides total estimated nuclear weapons spending by estimated inventory.

Figure 7. Estimated nuclear weapons spending per estimated warhead inventory

Data Tables

China stockpile estimates

Year	Estimated warheads	Source note
2015	260	Approximate historical open-source estimate; trend baseline
2020	350	Approximate open-source estimate; trend baseline
2021	400	Approximate open-source estimate; trend baseline
2022	410	Approximate open-source estimate; trend baseline
2023	500	Approximate open-source estimate; trend baseline
2024	500	SIPRI reports increase from 500 to up to 600 during 2024; this row uses 500 as January 2024 estimate
2025	600	SIPRI Yearbook 2025 and FAS/Bulletin estimate roughly/at least 600 warheads

Global nuclear force comparison

Country	Estimated warheads, 2025	Source note
Russia	4,309	SIPRI/FAS-style public estimate; rounded
United States	3,700	SIPRI/FAS-style public estimate; rounded
China	600	SIPRI/FAS/Bulletin public estimate; rounded
France	290	SIPRI/FAS-style public estimate; rounded
United Kingdom	225	SIPRI/FAS-style public estimate; rounded
India	180	SIPRI/FAS-style public estimate; rounded
Pakistan	170	SIPRI/FAS-style public estimate; rounded
Israel	90	Public estimate; Israel maintains opacity
North Korea	50	Public estimate with high uncertainty

Public Delivery-System Summary

Leg	System/category	Role in study	Public source note
Land-based	DF-31 family	ICBM force modernization indicator	FAS/Bulletin and CSIS public reporting
Land-based	DF-41	Road-mobile and silo-associated ICBM modernization indicator	FAS/Bulletin and CSIS public reporting
Land-based	DF-26	Regional dual-capable missile force indicator	FAS/Bulletin reports DF-26 appears to have replaced DF-21 in the nuclear role
Sea-based	Type 094 SSBN / JL-3 SLBM	Sea-based deterrent indicator	CSIS/FAS report Type 094 submarines refitted/equipped with JL-3; exact deployment details are uncertain
Air-based	H-6N bomber	Emerging/limited air-delivered nuclear role indicator	FAS/Bulletin reports reassigned operational nuclear mission to some bombers

China Defense and Nuclear Spending Inputs

N/A indicates that a reliable public China-specific estimate is not available in the current dataset.

Year	Official defense budget	Est. actual military spending	Est. nuclear weapons spending	Est. warheads
2023	N/A	N/A	$11.9B	500
2024	$232.0B	$318.0B	$12.5B	500
2025	$244.8B	$336.0B	N/A	600
2026	$276.7B	N/A	N/A	N/A

Derived Spending Metrics

Year	Nuclear share of official defense budget	Nuclear share of estimated actual spending	Nuclear spending / warhead inventory
2023	N/A	N/A	$23.8M
2024	5.39%	3.93%	$25.0M
2025	N/A	N/A	N/A
2026	N/A	N/A	N/A

Note: The spending-per-warhead inventory metric is not the cost of one warhead. It divides estimated nuclear weapons spending by estimated warhead inventory for comparison only.

Spending Source Notes

Year	Military spending source note	Nuclear spending source note
2023		ICAN estimate reported by The Guardian for 2023 nuclear weapons spending
2024	CSIS reports official defense budget around 2024 level; CSIS cites SIPRI estimate of 318 billion for 2024	ICAN estimate reported by Reuters for 2024 nuclear weapons spending
2025	CSIS reports 2025 official defense budget of 1.78 trillion yuan / 244.8 billion USD; SIPRI estimates 336 billion actual military expenditure	No reliable public ICAN-style China-specific nuclear weapons spending estimate available yet
2026	CSIS reports 2026 official defense budget of RMB 1.91 trillion / 276.7 billion USD; estimated actual spending not yet available	No reliable public ICAN-style China-specific nuclear weapons spending estimate available yet

Source Confidence Framework

Sources are rated based on institutional credibility, transparency of methodology, and usefulness for public nuclear-force analysis. The table below is a compact summary; full URLs and notes follow separately.

Source ID	Source name	Organization	Year	Type	Confidence	Used for
SIPRI_2025	SIPRI Yearbook 2025	SIPRI	2025	Research institute	High	Stockpile estimate and global nuclear force context
FAS_BULLETIN_2025	Chinese Nuclear Weapons 2025	Federation of American Scientists / Bulletin of the Atomic Scientists	2025	Expert estimate	High	China stockpile estimate and delivery-system modernization
CSIS_CHINAPOWER	How is China Modernizing its Nuclear Forces?	CSIS ChinaPower	2025	Think tank analysis	Medium-High	Delivery-system modernization and strategic context
DOD_CMPR_2025	Military and Security Developments Involving the People's Republic of China 2025	U.S. Department of Defense	2025	Government report	Medium-High	2030 nuclear projection and defense-spending caveats
CSIS_MILEX_2026	What Does China Really Spend on its Military?	CSIS ChinaPower	2026	Think tank analysis	High	Official defense budget and spending transparency caveats
CSIS_GOV_BUDGET_2026	Making Sense of China's Government Budget	CSIS ChinaPower	2026	Think tank analysis	High	China government budget context and official defense budget
MOF_BUDGET_2026	Report on the Execution of the Central and Local Budgets for 2025 and on the Draft Central and Local Budgets for 2026	Ministry of Finance of the People's Republic of China	2026	Government budget report	High	Official China central government national defense budget figures
SIPRI_MILEX_2025	Global Military Spending Rise Continues	SIPRI	2026	Research institute	High	Estimated actual Chinese military expenditure for 2025
ICAN_NUCLEAR_SPENDING_2024	Global Nuclear Arms Spending Up 11% in 2024	ICAN / Reuters	2025	NGO estimate reported by wire service	Medium-High	Estimated China nuclear weapons spending for 2024
ICAN_NUCLEAR_SPENDING_2023	Global Spending on Nuclear Weapons Up 13%	ICAN / The Guardian	2024	NGO estimate reported by news source	Medium	Estimated China nuclear weapons spending for 2023
BBVA_FISCAL_BUDGET_2025	Decoding China's Multi-Layered Fiscal Budget	BBVA Research	2025	Economic research report	Medium-High	China fiscal-budget structure and budget interpretation context
CRS_IF11737_2025	China: Nuclear and Missile Proliferation	Congressional Research Service	2025	Government research report	Medium-High	Nonproliferation context and China missile/nuclear export-control background

Source URLs and Notes

SIPRI_2025: SIPRI Yearbook 2025
URL: https://www.sipri.org/yearbook/2025
Notes: Useful for broad nuclear force estimates and trend framing

FAS_BULLETIN_2025: Chinese Nuclear Weapons 2025
URL: https://fas.org/wp-content/uploads/2025/03/Chinese-nuclear-weapons-2025.pdf
Notes: Detailed public nuclear force estimate with transparent caveats; useful for China's estimated stockpile and delivery-system modernization

CSIS_CHINAPOWER: How is China Modernizing its Nuclear Forces?
URL: https://chinapower.csis.org/china-nuclear-weapons/
Notes: Useful for explaining China's triad development and modernization

DOD_CMPR_2025: Military and Security Developments Involving the People's Republic of China 2025
URL: https://media.defense.gov/2025/Dec/23/2003849070/-1/-1/1/ANNUAL-REPORT-TO-CONGRESS-MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2025.PDF
Notes: Useful for official U.S. assessment that China remains on track for over 1,000 warheads by 2030 and that announced defense spending understates total defense spending

CSIS_MILEX_2026: What Does China Really Spend on its Military?
URL: https://chinapower.csis.org/military-spending/
Notes: Useful for official budget figures and discussion of underreporting

CSIS_GOV_BUDGET_2026: Making Sense of China's Government Budget
URL: https://chinapower.csis.org/making-sense-of-chinas-government-budget/
Notes: Useful for 2026 official defense budget, fiscal context, and explanation that official figures do not capture total military spending

Source URLs and Notes - continued

MOF_BUDGET_2026: Report on the Execution of the Central and Local Budgets for 2025 and on the Draft Central and Local Budgets for 2026
URL: uploaded_docx
Notes: Official budget report useful for 2025 and 2026 national defense expenditure; does not provide nuclear-specific spending

SIPRI_MILEX_2025: Global Military Spending Rise Continues
URL: https://www.sipri.org/media/press-release/2026/global-military-spending-rise-continues-european-and-asian-expenditures-surge
Notes: SIPRI estimated China spent 336 billion USD on its military in 2025

ICAN_NUCLEAR_SPENDING_2024: Global Nuclear Arms Spending Up 11% in 2024
URL: https://www.reuters.com/world/china/global-nuclear-arms-spending-up-11-2024-campaign-group-says-2025-06-13/
Notes: Reuters reports ICAN estimate that China spent 12.5 billion USD on nuclear weapons in 2024

ICAN_NUCLEAR_SPENDING_2023: Global Spending on Nuclear Weapons Up 13%
URL: https://www.theguardian.com/world/article/2024/jun/17/global-spending-on-nuclear-weapons-up-13-in-record-rise
Notes: The Guardian reports ICAN estimate that China spent 11.9 billion USD on nuclear weapons in 2023

BBVA_FISCAL_BUDGET_2025: Decoding China's Multi-Layered Fiscal Budget
URL: https://www.bbvaresearch.com/wp-content/uploads/2025/10/China_Decoding_Chinas_Multi_Layered_Fiscal_Budget.pdf
Notes: Useful for explaining China's multi-layered fiscal system and why headline budget figures require careful interpretation

Source URLs and Notes - continued

CRS_IF11737_2025: China: Nuclear and Missile Proliferation
URL: https://crsreports.congress.gov/product/pdf/IF/IF11737
Notes: Useful for background on China-related nuclear and missile proliferation concerns, export-control issues, and U.S. sanctions; not a direct stockpile or nuclear spending estimate

Limitations

• All nuclear stockpile numbers are estimates. China does not publish a detailed nuclear inventory, and outside analysts rely on a mix of official statements, satellite imagery, budget signals, public military reporting, and expert judgment. This means the exact number of warheads, deployed systems, and operational readiness levels remain uncertain.

• China does not publish a separate nuclear weapons budget, and even its official defense budget does not fully capture total military-related spending. As a result, the study uses proxy variables and outside estimates. These should be interpreted as directional indicators rather than exact accounting data.

• The spending-per-warhead figure should not be interpreted as the cost of one nuclear warhead. Nuclear weapons spending can include warheads, delivery systems, submarines, missile infrastructure, command and control, personnel, research, production facilities, and modernization.

Selected Source Base

• SIPRI Yearbook 2025, World Nuclear Forces

• SIPRI military expenditure reporting

• Federation of American Scientists / Bulletin of the Atomic Scientists, Chinese Nuclear Weapons 2025

• CSIS ChinaPower, How is China Modernizing its Nuclear Forces?

• CSIS ChinaPower, What Does China Really Spend on its Military?

• CSIS ChinaPower, Making Sense of China's Government Budget

• Ministry of Finance of the People's Republic of China, Report on the Execution of the Central and Local Budgets for 2025 and on the Draft Central and Local Budgets for 2026

• U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China 2025

• ICAN nuclear weapons spending estimates reported by Reuters and The Guardian

• BBVA Research, Decoding China's Multi-Layered Fiscal Budget

• Congressional Research Service, China: Nuclear and Missile Proliferation